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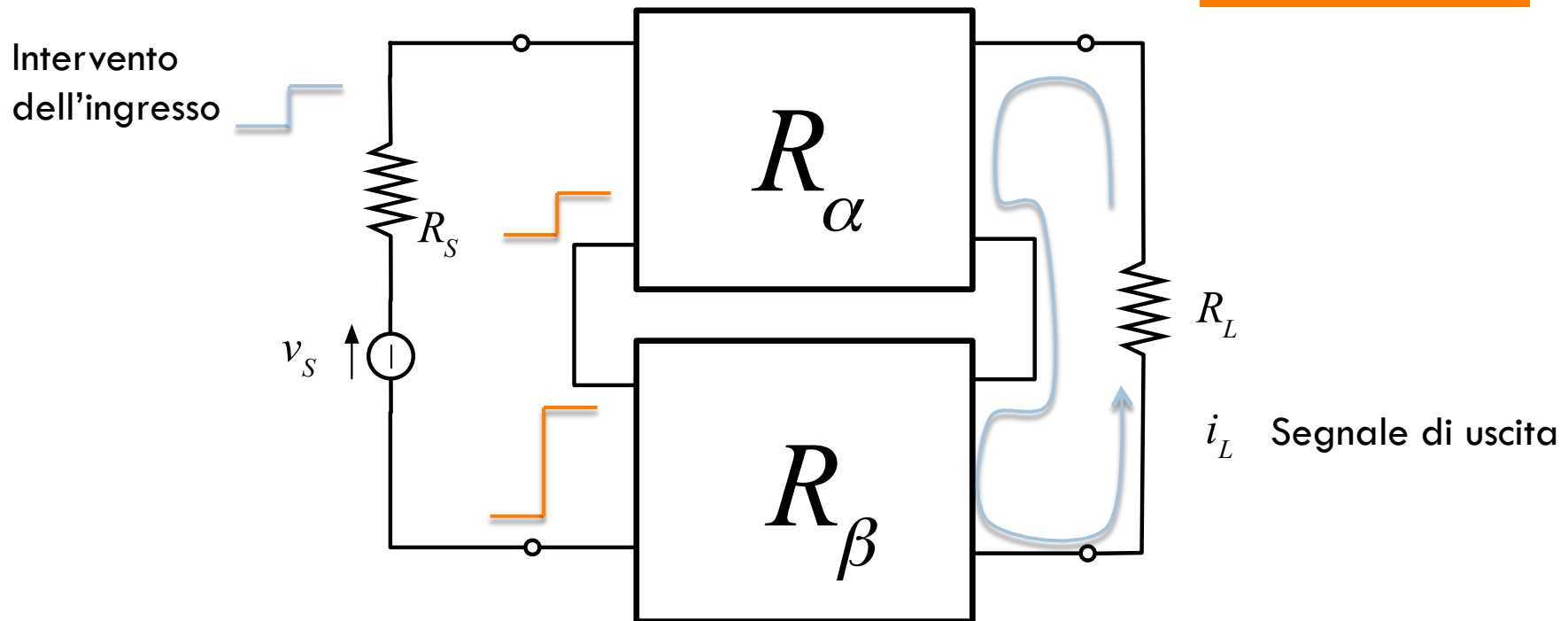
FEEDBACK SYSTEM 2

Associate professor Stefano Saggin
Lezione di complementi di elettronica

Retroazione serie serie

- Accesso ad un ramo della circuito retroazionato
l'uscita è su un ramo della retrazione

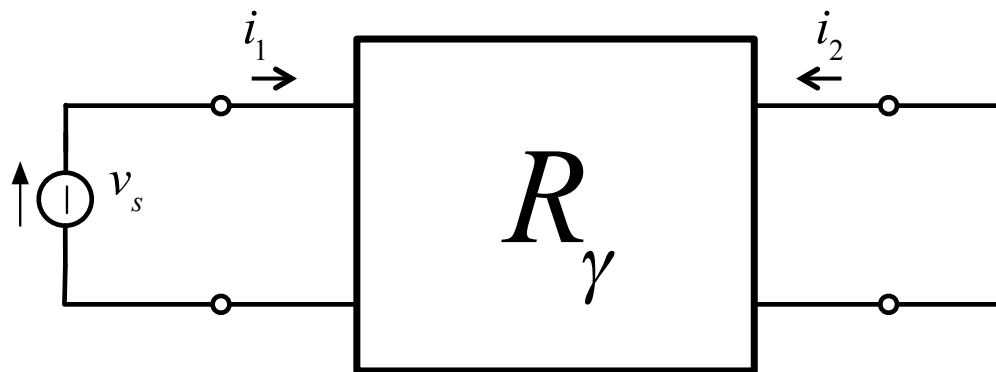
Effetto della
retroazione negativa



Retroazione serie serie

□ Calcolo del risultato

$$R_\gamma = R_\alpha + R_\beta + \begin{bmatrix} R_S & 0 \\ 0 & R_L \end{bmatrix} = \begin{bmatrix} r_{11}^\alpha + r_{11}^\beta + R_S & r_{12}^\alpha + r_{12}^\beta \\ r_{21}^\alpha + r_{21}^\beta & r_{22}^\alpha + r_{22}^\beta + R_L \end{bmatrix}$$



$$\begin{bmatrix} v_s \\ 0 \end{bmatrix} = R_\gamma \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$


$$R_\gamma^{-1} \begin{bmatrix} v_s \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

Retroazione serie serie

□ Soluzione

$$\frac{1}{\det(R_\gamma)} \begin{bmatrix} r_{22}^\alpha + r_{22}^\beta + R_L & -(r_{12}^\alpha + r_{12}^\beta) \\ -(r_{21}^\alpha + r_{21}^\beta) & r_{11}^\alpha + r_{11}^\beta + r_s \end{bmatrix} \begin{bmatrix} v_s \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

□ Analisi del primo trasferimento

Termine dominante G_D

Termine parassita G_P

$$\frac{i_2}{v_s} = \frac{-(r_{21}^\alpha + r_{21}^\beta)}{\det R_\gamma} = \frac{-r_{21}^\alpha}{\det R_\gamma} + \frac{-r_{21}^\beta}{\det R_\gamma}$$

Trasferimento

□ Trasferimento

Termine
dominante G_D

Termine parassita G_p

$$\frac{i_2}{v_S} = \frac{-(r_{21}^\alpha + r_{21}^\beta)}{\det R_\gamma} = \frac{-r_{21}^\alpha}{\det R_\gamma} + \frac{-r_{21}^\beta}{\det R_\gamma}$$

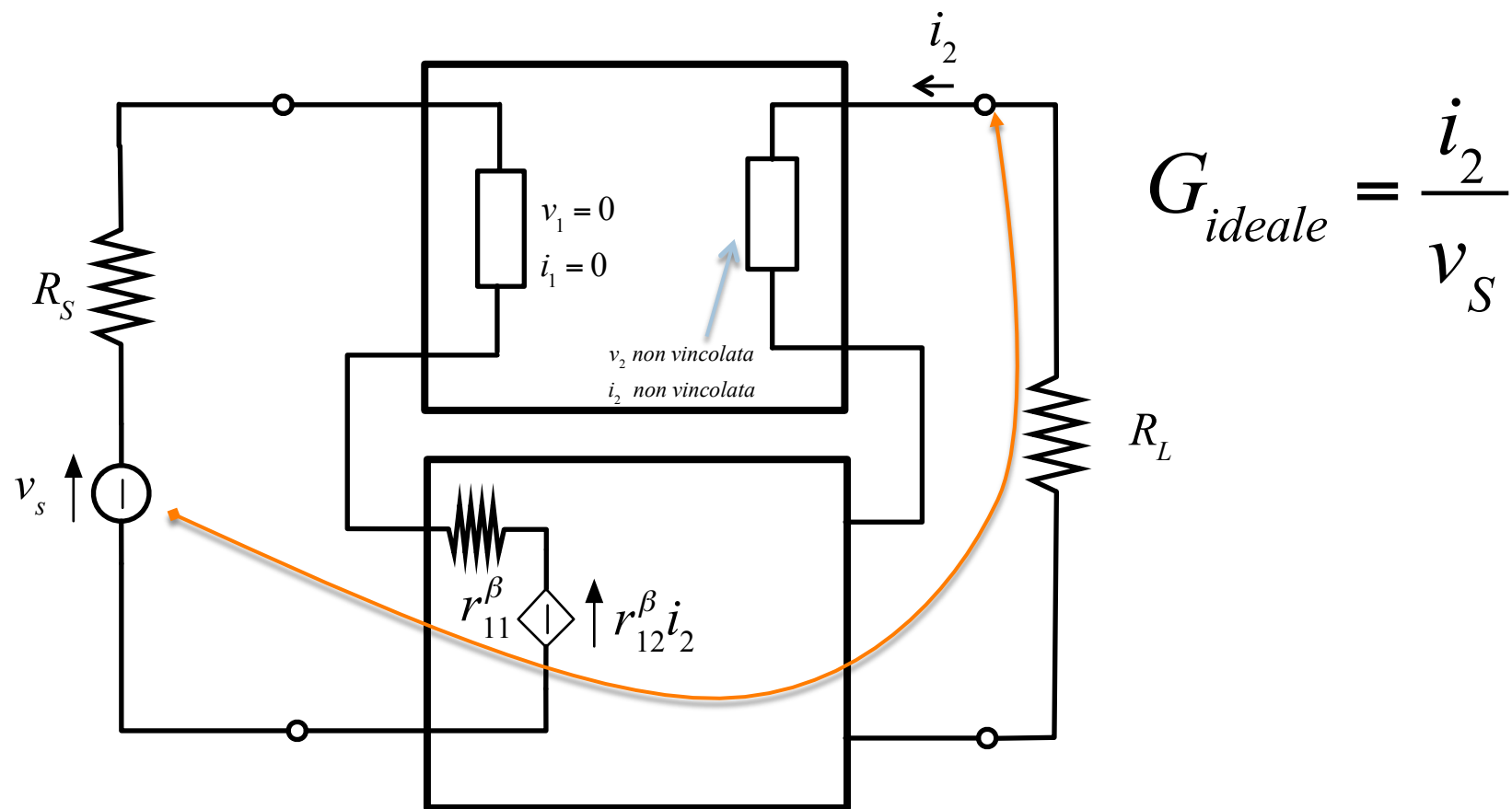
$$G_{reale} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} + \frac{G_{diretto}}{1 - G_{loop}}$$

$$R_\alpha = \begin{bmatrix} R_{in} & \cong 0 \\ R & R_{out} \end{bmatrix}$$

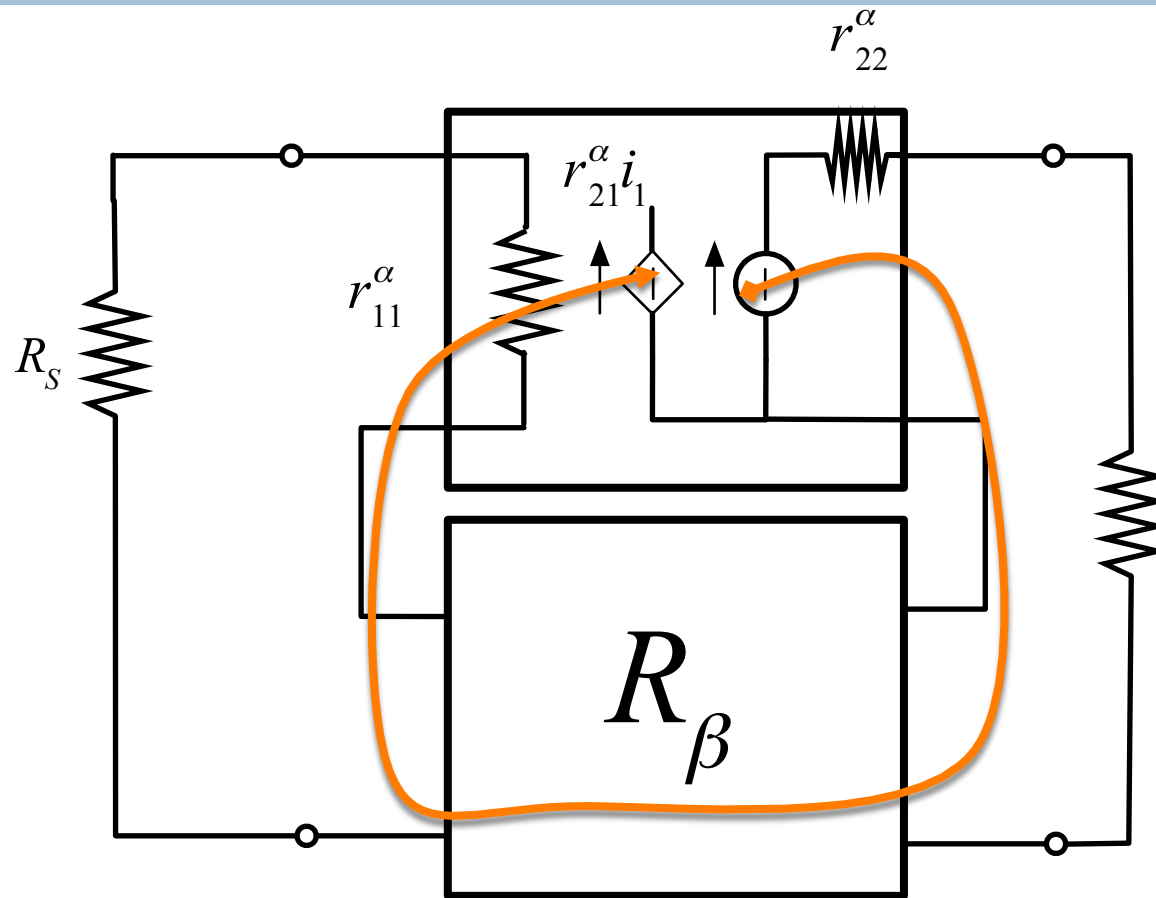
$$G_{ideale} = \lim_{R \rightarrow -\infty} G_D = \lim_{R \rightarrow -\infty} \frac{-R}{(r_{11}^\alpha + r_{11}^\beta + R_S)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)(R + r_{21}^\beta)} = \frac{1}{(r_{12}^\alpha + r_{12}^\beta)} \cong \frac{1}{r_{12}^\beta}$$

$$G_D = G_{ideale} \frac{1}{1 + \frac{r_{21}^\beta (r_{12}^\alpha + r_{12}^\beta) - (r_{11}^\alpha + r_{11}^\beta + R_S)(r_{22}^\alpha + r_{22}^\beta + R_L)}{(r_{12}^\alpha + r_{12}^\beta) r_{21}^\alpha}} = G_{ideale} \frac{1}{1 - G_{loop}^{-1}}$$

G_{ideale} nel circuito



G_{loop} nel circuito



$$G_{\text{loop}} = \frac{-\left(r_{12}^\alpha + r_{12}^\beta\right)r_{21}^\alpha}{r_{21}^\beta\left(r_{12}^\alpha + r_{12}^\beta\right) - \left(r_{11}^\alpha + r_{11}^\beta + R_S\right)\left(r_{22}^\alpha + r_{22}^\beta + R_L\right)}$$

Analisi del secondo termine

□ Analisi del secondo termine

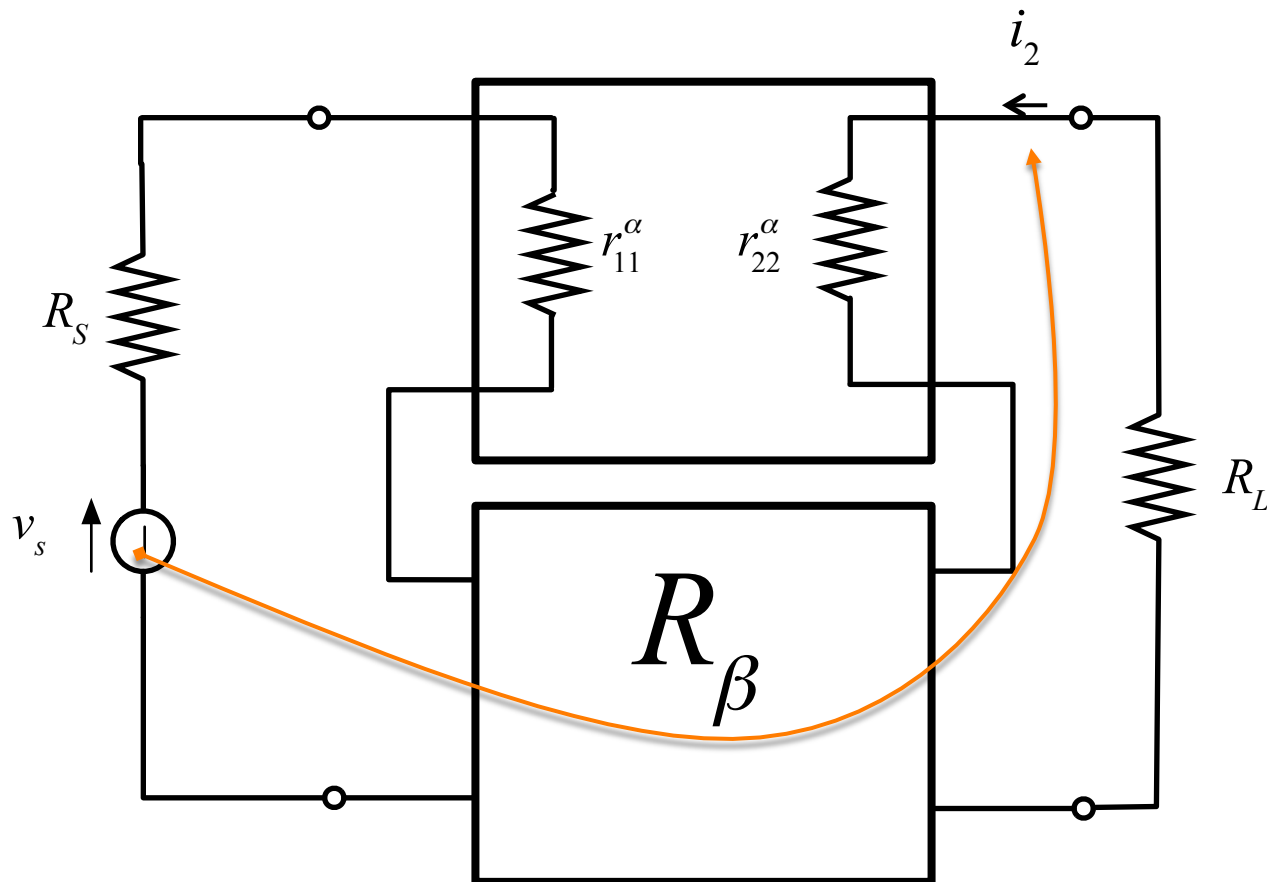
Guadagno diretto

$$G_P = \frac{r_{21}^\beta}{r_{21}^\beta (r_{12}^\alpha + r_{12}^\beta) - (r_{11}^\alpha + r_{11}^\beta + R_S)(r_{22}^\alpha + r_{22}^\beta + R_L)} \frac{1}{1 - G_{loop}}$$

$$G_{diretto} = \frac{r_{21}^\beta}{r_{21}^\beta (r_{12}^\alpha + r_{12}^\beta) - (r_{11}^\alpha + r_{11}^\beta + R_S)(r_{22}^\alpha + r_{22}^\beta + R_L)}$$

$$G_P = G_{diretto} \frac{1}{1 - G_{loop}}$$

G_{diretto} nel circuito



$$G_{\text{diretto}} = \frac{i_2}{v_s}$$

Analisi del secondo trasferimento

□ Impedenza di ingresso

$$\frac{1}{\det(R_\gamma)} \begin{bmatrix} r_{22}^\alpha + r_{22}^\beta + R_L & -(r_{12}^\alpha + r_{12}^\beta) \\ -(r_{21}^\alpha + r_{21}^\beta) & r_{11}^\alpha + r_{11}^\beta + R_S \end{bmatrix} \begin{bmatrix} v_S \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

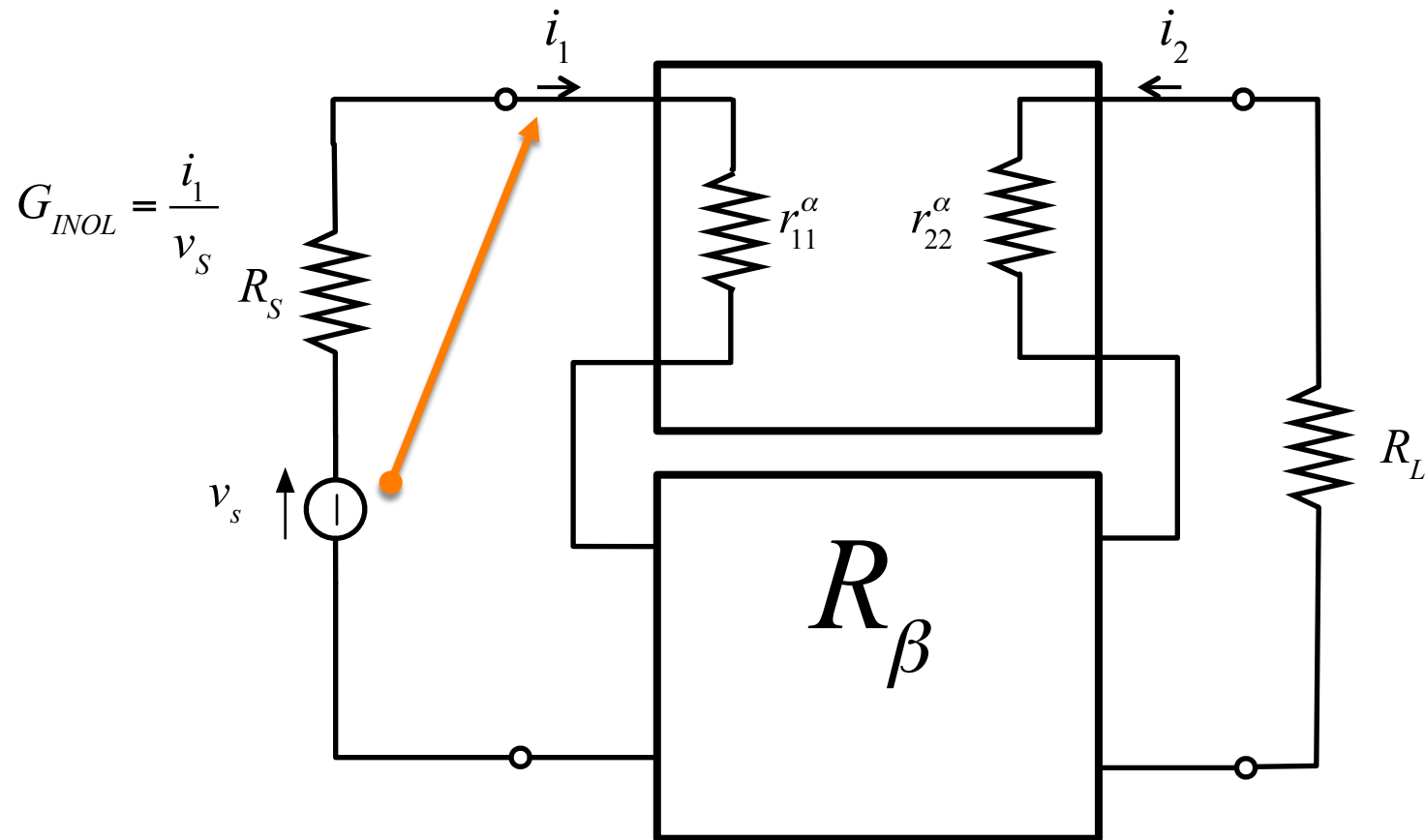
$$\frac{i_1}{v_S} = \frac{(r_{22}^\alpha + r_{22}^\beta + R_L)}{\det R_\gamma} = \frac{r_{22}^\alpha + r_{22}^\beta + R_L}{(r_{11}^\alpha + r_{11}^\beta + R_S)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)(r_{21}^\alpha + r_{21}^\beta)}$$

$$G_{IN} = \frac{r_{22}^\alpha + r_{22}^\beta + R_L}{(r_{11}^\alpha + r_{11}^\beta + R_S)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)r_{21}^\beta} \cdot \frac{1}{1 - \frac{(r_{12}^\alpha + r_{12}^\beta)r_{21}^\alpha}{(r_{11}^\alpha + r_{11}^\beta + R_S)(r_{22}^\alpha + r_{22}^\beta + R_L) - (r_{12}^\alpha + r_{12}^\beta)r_{21}^\beta}}$$

Conduttanza anello aperto

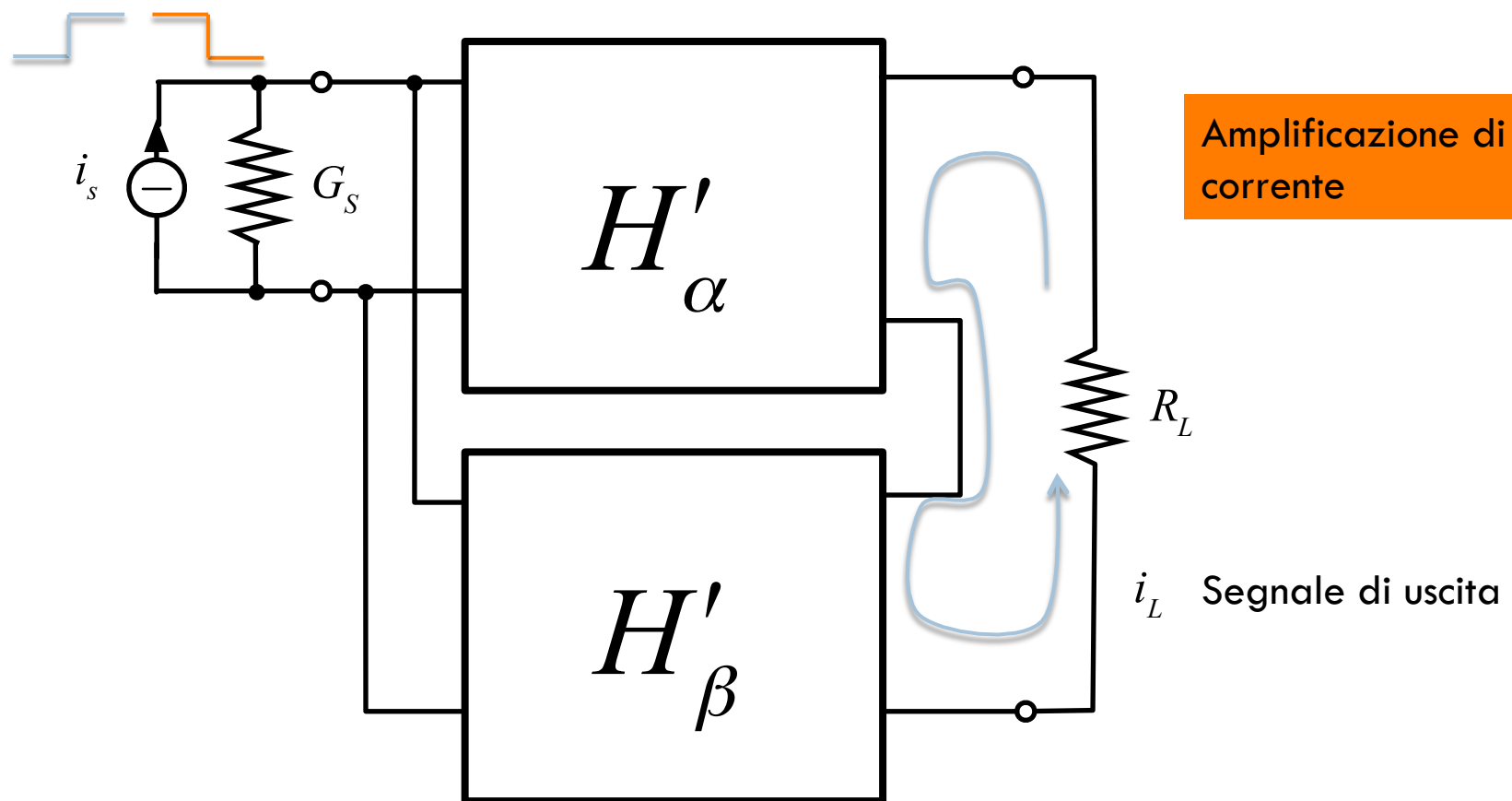
$$G_{IN} = G_{INOL} \frac{1}{1 - G_{loop}}$$

$G_{\text{anello aperto}}$ nel circuito



Retroazione parallelo serie

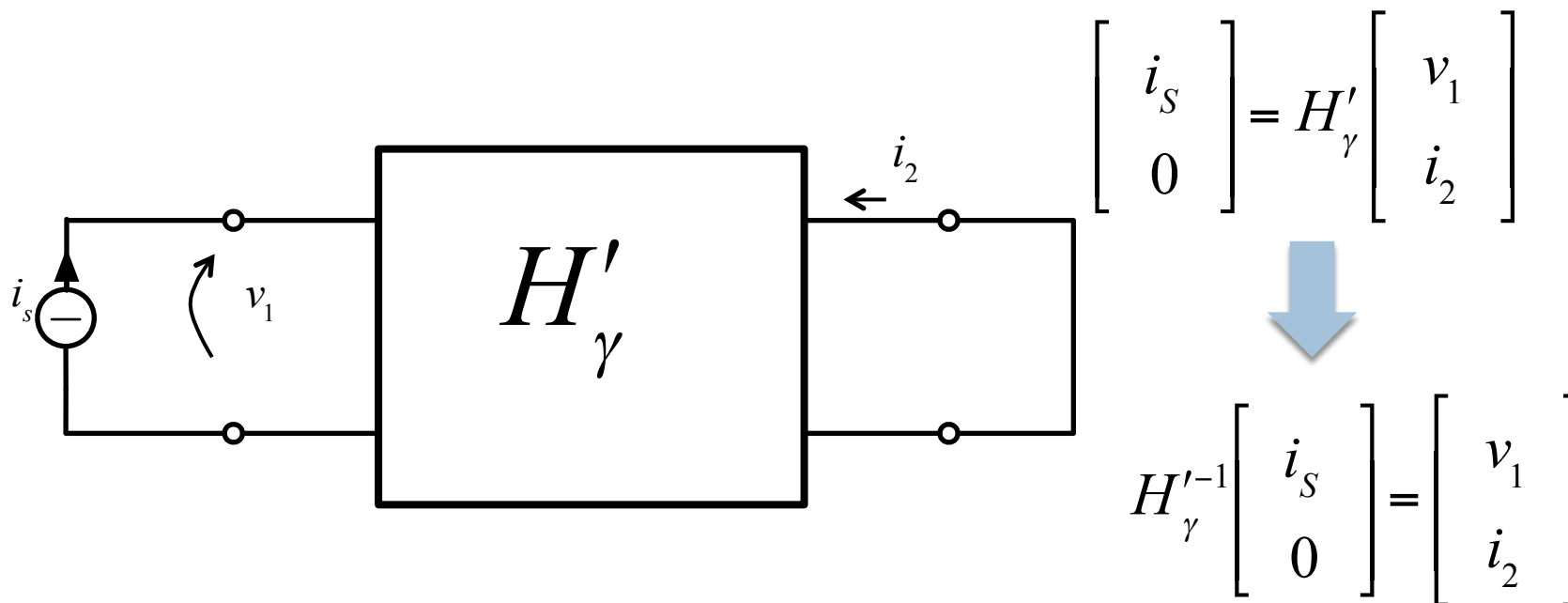
- Accesso ad un nodo del circuito retroazionato
l'uscita è su un ramo della retroazione



Retroazione parallelo parallelo

□ Calcolo del risultato

$$H'_\gamma = H'_\alpha + H'_\beta + \begin{bmatrix} G_S & 0 \\ 0 & R_L \end{bmatrix} = \begin{bmatrix} h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S & h'_{12}{}^\alpha + h'_{12}{}^\beta \\ h'_{21}{}^\alpha + h'_{21}{}^\beta & h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L \end{bmatrix}$$



Retroazione parallelo serie

□ Soluzione

$$\frac{1}{\det(H'_\gamma)} \begin{bmatrix} h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L & -(h'_{12}{}^\alpha + h'_{12}{}^\beta) \\ -(h'_{21}{}^\alpha + h'_{21}{}^\beta) & h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S \end{bmatrix} \begin{bmatrix} i_S \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$

□ Analisi del primo trasferimento

Termine dominante A_{ID}

$$\frac{i_2}{i_S} = \frac{-(h'_{21}{}^\alpha + h'_{21}{}^\beta)}{\det H'_\gamma} = \frac{-h'_{21}{}^\alpha}{\det H'_\gamma} + \frac{-h'_{21}{}^\beta}{\det H'_\gamma}$$

Termine parassita A_{IP}

Trasferimento

□ Trasferimento

Termine
dominante A_{ID}

Termine parassita A_{IP}

$$\frac{i_2}{i_s} = \frac{-(h'_{21} + h'_{21})}{\det H'_\gamma} = \frac{-h'_{21}}{\det H'_\gamma} + \frac{-h'_{21}}{\det H'_\gamma}$$

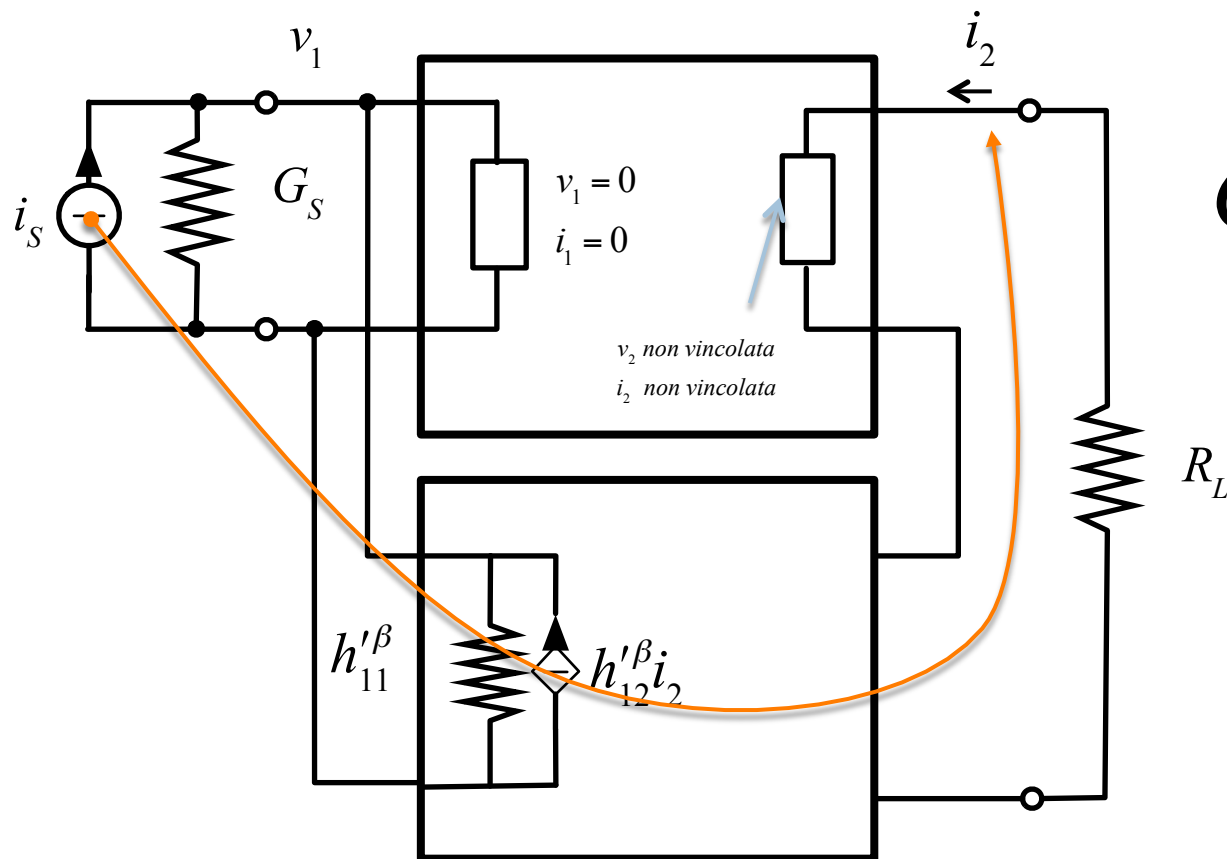
$$G_{reale} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} + \frac{G_{diretto}}{1 - G_{loop}}$$

$$H'_\alpha = \begin{bmatrix} 1/R_{in} & \cong 0 \\ A_v & R_{out} \end{bmatrix}$$

$$G_{ideale} = \lim_{A_v \rightarrow -\infty} A_{ID} = \lim_{A_v \rightarrow -\infty} \frac{-A_v}{(h'_{11} + h'_{11} + G_s)(h'_{22} + h'_{22} + R_L) - (h'_{12} + h'_{12})(A_v + h'_{21})} = \frac{1}{(h'_{12} + h'_{12})} \cong \frac{1}{h'_{12}}$$

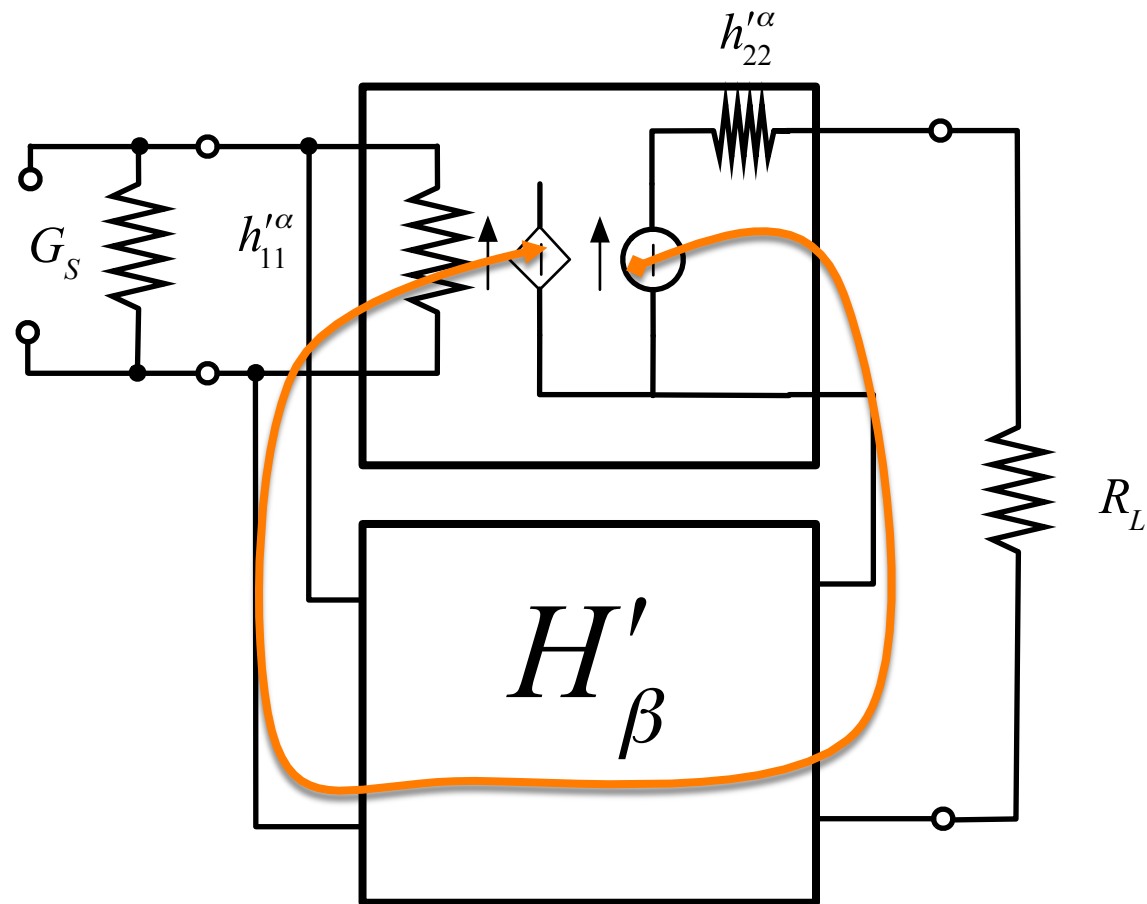
$$G_D = G_{ideale} \frac{1}{1 + \frac{h'_{21}(h'_{12} + h'_{12}) - (h'_{11} + h'_{11} + G_s)(h'_{22} + h'_{22} + R_L)}{(h'_{12} + h'_{12})h'_{21}}} = G_{ideale} \frac{1}{1 - G_{loop}^{-1}}$$

G_{ideale} nel circuito



$$G_{ideale} = \frac{i_2}{i_S}$$

G_{loop} nel circuito



$$G_{loop} = \frac{-(h'_{12}{}^{\alpha} + h'_{12}{}^{\beta})h'_{21}{}^{\alpha}}{h'_{21}{}^{\beta}(h'_{12}{}^{\alpha} + h'_{12}{}^{\beta}) - (h'_{11}{}^{\alpha} + h'_{11}{}^{\beta} + G_S)(h'_{22}{}^{\alpha} + h'_{22}{}^{\beta} + R_L)}$$

Analisi del secondo termine

□ Analisi del secondo termine

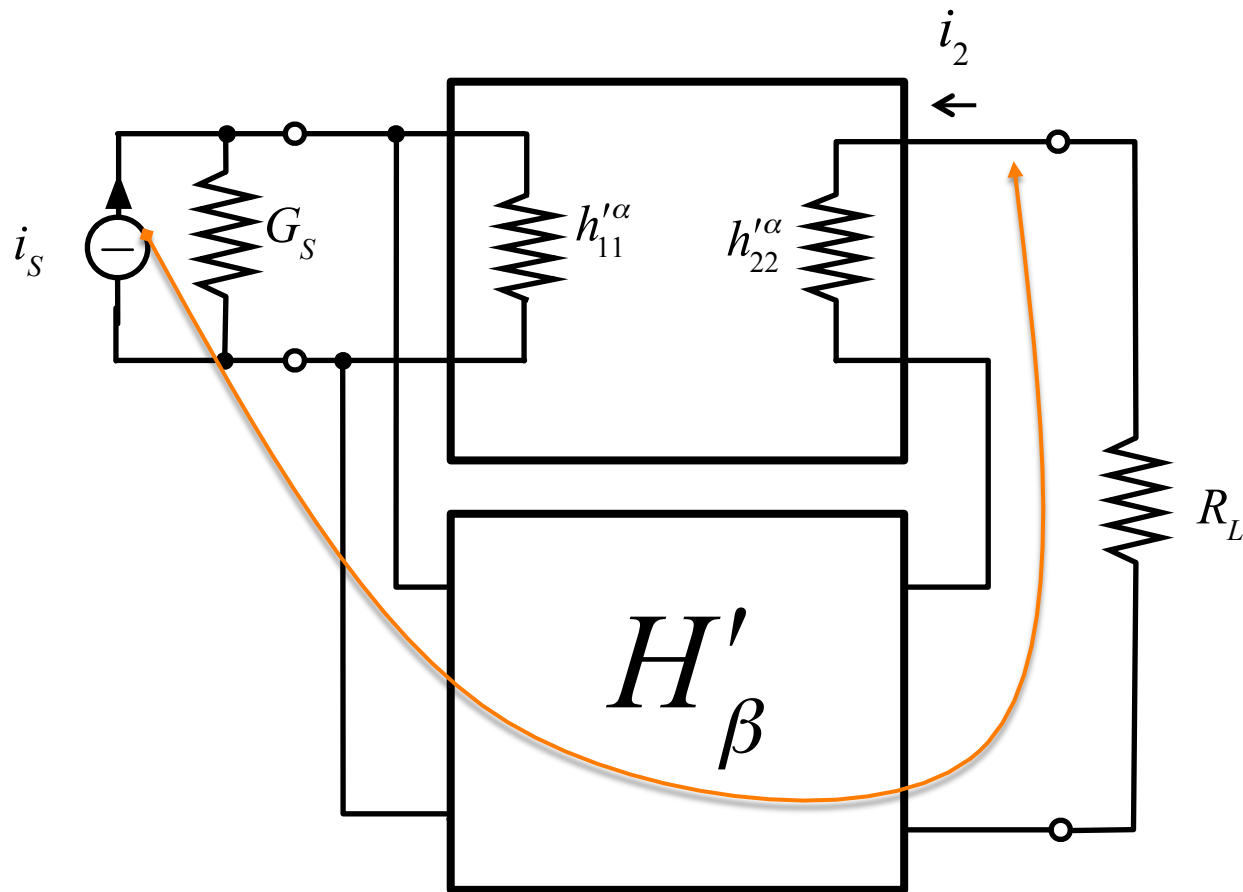
Guadagno diretto

$$A_{IP} = \frac{h'_{21}{}^{\beta}}{h'_{21}{}^{\beta}(h'_{12}{}^{\alpha} + h'_{12}{}^{\beta}) - (h'_{11}{}^{\alpha} + h'_{11}{}^{\beta} + G_S)(h'_{22}{}^{\alpha} + h'_{22}{}^{\beta} + R_L)} \frac{1}{1 - G_{loop}}$$

$$G_{diretto} = \frac{h'_{21}{}^{\beta}}{h'_{21}{}^{\beta}(h'_{12}{}^{\alpha} + h'_{12}{}^{\beta}) - (h'_{11}{}^{\alpha} + h'_{11}{}^{\beta} + G_S)(h'_{22}{}^{\alpha} + h'_{22}{}^{\beta} + R_L)}$$

$$G_P = G_{diretto} \frac{1}{1 - G_{loop}}$$

G_{diretto} nel circuito



$$G_{\text{diretto}} = \frac{i_2}{i_s}$$

Analisi del secondo trasferimento

□ Impedenza di ingresso

$$\frac{1}{\det(H'_\gamma)} \begin{bmatrix} h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L & -(h'_{12}{}^\alpha + h'_{12}{}^\beta) \\ -(h'_{21}{}^\alpha + h'_{21}{}^\beta) & h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S \end{bmatrix} \begin{bmatrix} i_S \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$

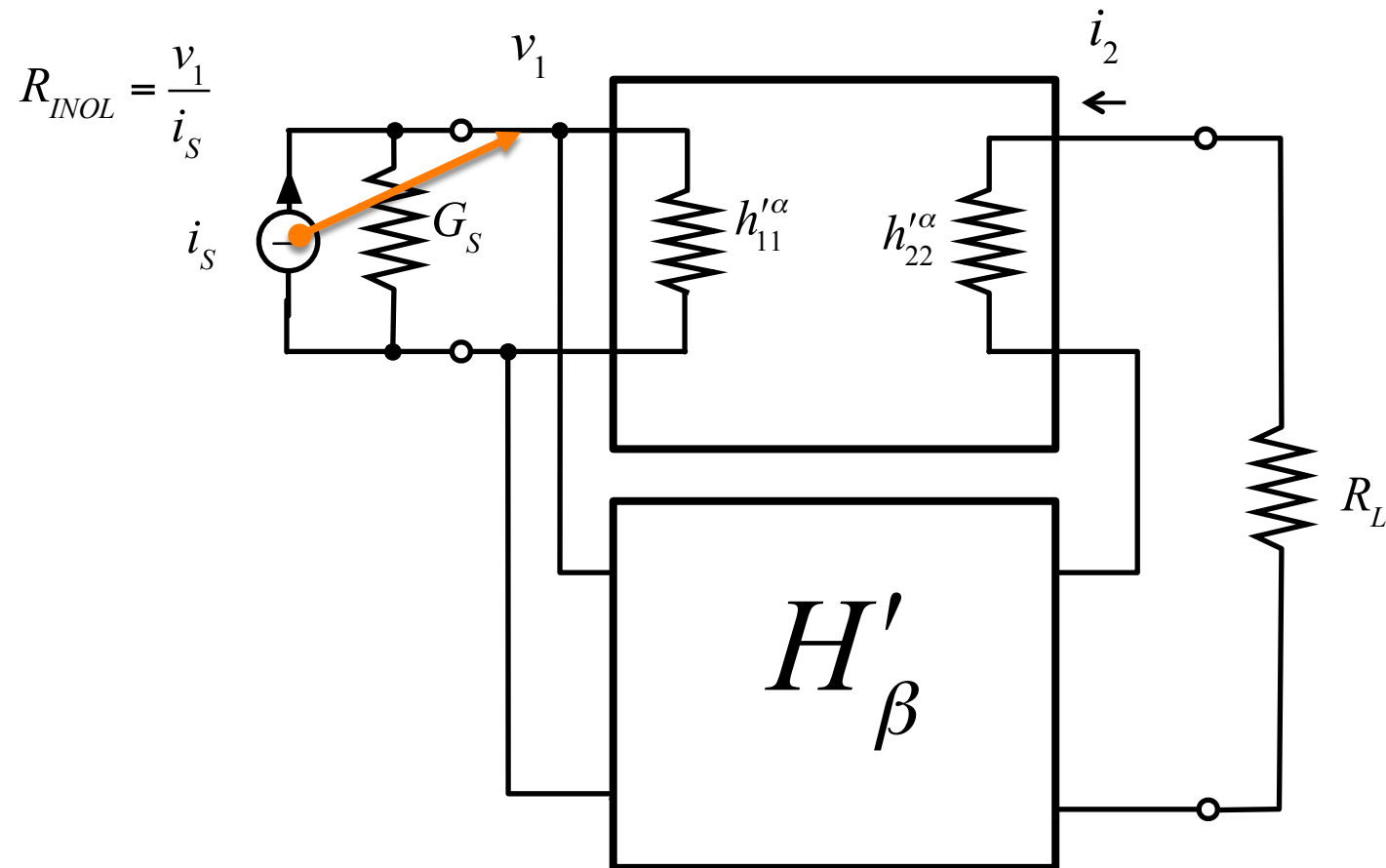
$$\frac{v_1}{i_S} = \frac{(h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L)}{\det H'_\gamma} = \frac{h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L}{(h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S)(h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L) - (h'_{12}{}^\alpha + h'_{12}{}^\beta)(h'_{21}{}^\alpha + h'_{21}{}^\beta)}$$

$$R_{IN} = \frac{h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L}{(h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S)(h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L) - (h'_{12}{}^\alpha + h'_{12}{}^\beta)h'_{21}{}^\beta} - \frac{1}{1 - \frac{(h'_{12}{}^\alpha + h'_{12}{}^\beta)h'_{21}{}^\alpha}{(h'_{11}{}^\alpha + h'_{11}{}^\beta + G_S)(h'_{22}{}^\alpha + h'_{22}{}^\beta + R_L) - (h'_{12}{}^\alpha + h'_{12}{}^\beta)h'_{21}{}^\beta}}$$

Resistenza anello aperto

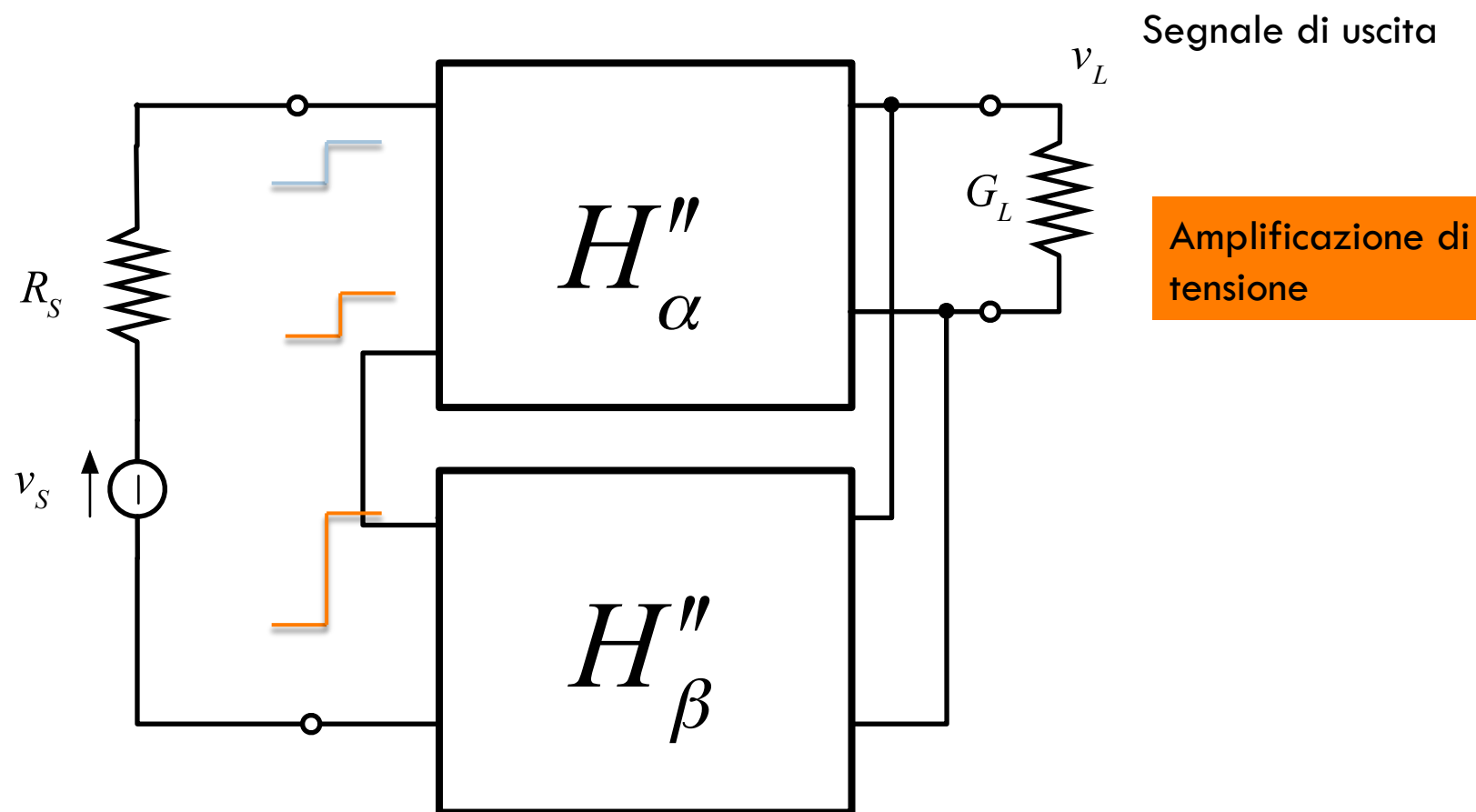
$$R_{IN} = R_{INOL} \frac{1}{1 - G_{loop}}$$

$G_{\text{anello aperto}}$ nel circuito



Retroazione serie parallelo

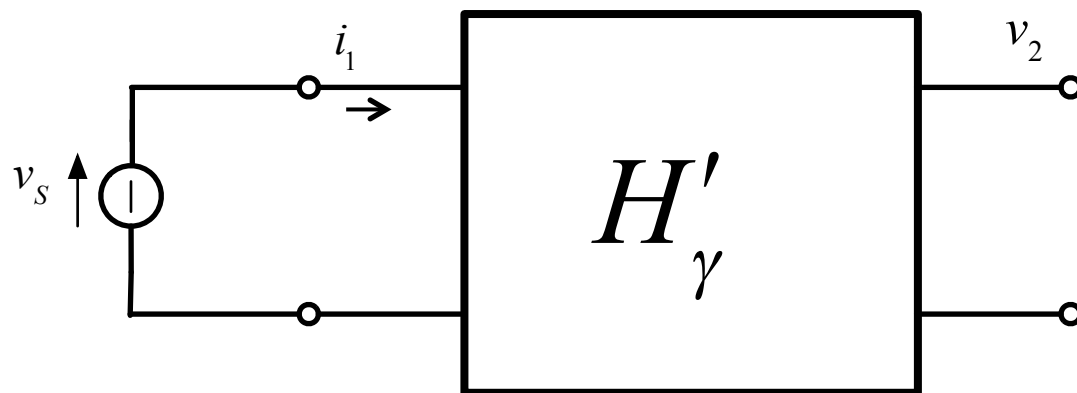
- Accesso ad un nodo del circuito retroazionato
l'uscita è su un ramo della retroazione



Retroazione parallelo parallelo

□ Calcolo del risultato

$$H''_{\gamma} = H''_{\alpha} + H''_{\beta} + \begin{bmatrix} R_S & 0 \\ 0 & G_L \end{bmatrix} = \begin{bmatrix} h''^{\alpha}_{11} + h''^{\beta}_{11} + R_S & h''^{\alpha}_{12} + h''^{\beta}_{12} \\ h''^{\alpha}_{21} + h''^{\beta}_{21} & h''^{\alpha}_{22} + h''^{\beta}_{22} + G_L \end{bmatrix}$$



$$\begin{bmatrix} v_S \\ 0 \end{bmatrix} = H''_{\gamma} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

↓

$$H''_{\gamma}^{-1} \begin{bmatrix} v_S \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

Retroazione parallelo serie

□ Soluzione

$$\frac{1}{\det(H''_\gamma)} \begin{bmatrix} h''_{22} + h''_{22} + G_L & -(h''_{12} + h''_{12}) \\ -(h''_{21} + h''_{21}) & h''_{11} + h''_{11} + R_S \end{bmatrix} \begin{bmatrix} v_S \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

□ Analisi del primo trasferimento

Termine dominante A_{VD}

$$\frac{v_2}{v_S} = \frac{-(h''_{21} + h''_{21})}{\det H''_\gamma} = \frac{-h''_{21}}{\det H''_\gamma} + \frac{-h''_{21}}{\det H''_\gamma}$$

Termine parassita A_{VP}

Trasferimento

□ Trasferimento

Termine
dominante A_{VD}

Termine parassita A_{VP}

$$\frac{v_2}{v_S} = \frac{-(h_{21}''^{\alpha} + h_{21}''^{\beta})}{\det H_{\gamma}''} = \frac{-h_{21}''^{\alpha}}{\det H_{\gamma}''} + \frac{-h_{21}''^{\beta}}{\det H_{\gamma}''}$$

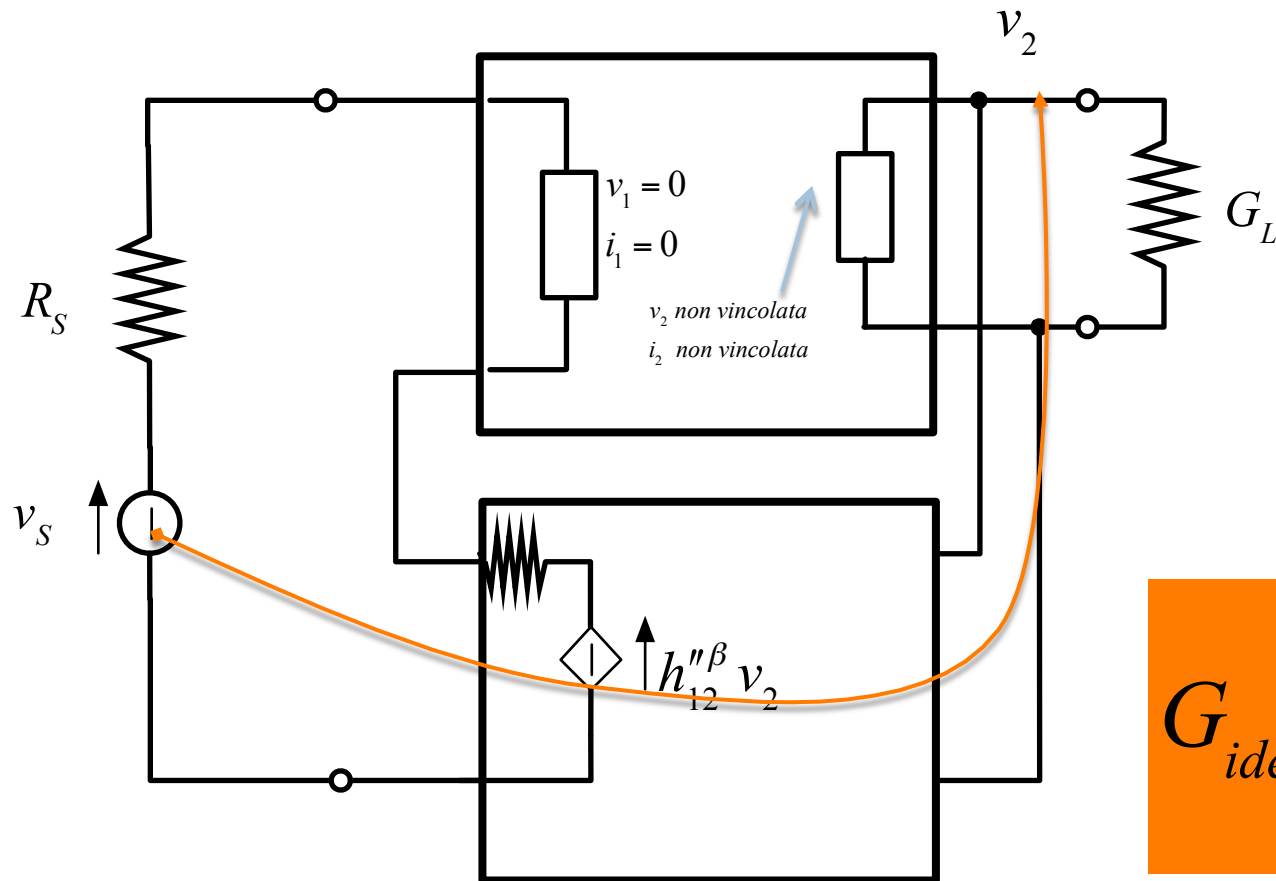
$$G_{reale} = \frac{G_{ideale}}{1 - G_{loop}^{-1}} + \frac{G_{diretto}}{1 - G_{loop}}$$

$$H_{\alpha}'' = \begin{bmatrix} R_{in} & \cong 0 \\ A_I & 1/R_{out} \end{bmatrix}$$

$$G_{ideale} = \lim_{A_I \rightarrow -\infty} A_{VD} = \lim_{A_I \rightarrow -\infty} \frac{-A_I}{(h_{11}''^{\alpha} + h_{11}''^{\beta} + R_S)(h_{22}''^{\alpha} + h_{22}''^{\beta} + G_L) - (h_{12}''^{\alpha} + h_{12}''^{\beta})(A_I + h_{21}''^{\beta})} = \frac{1}{(h_{12}''^{\alpha} + h_{12}''^{\beta})} \cong \frac{1}{h_{12}''^{\beta}}$$

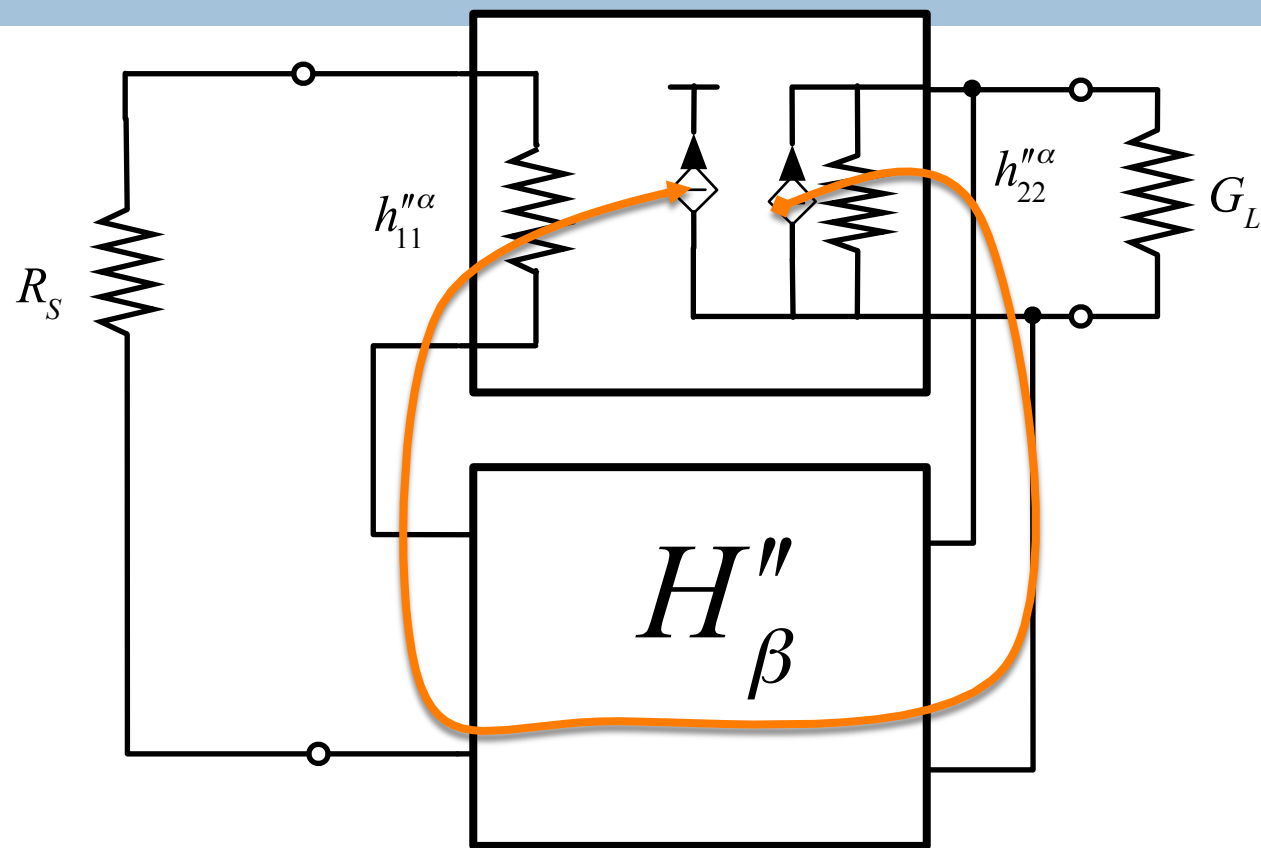
$$A_{VD} = G_{ideale} \frac{1}{1 + \frac{h_{21}''^{\beta}(h_{12}''^{\alpha} + h_{12}''^{\beta}) - (h_{11}''^{\alpha} + h_{11}''^{\beta} + R_S)(h_{22}''^{\alpha} + h_{22}''^{\beta} + G_L)}{(h_{12}''^{\alpha} + h_{12}''^{\beta})h_{21}''^{\alpha}}} = G_{ideale} \frac{1}{1 - G_{loop}^{-1}}$$

G_{ideale} nel circuito



$$G_{ideale} = \frac{v_2}{v_S}$$

G_{loop} nel circuito



$$G_{loop} = \frac{-\left(h''_{12}{}^{\alpha} + h''_{12}{}^{\beta}\right)h''_{21}{}^{\alpha}}{h''_{21}{}^{\beta}\left(h''_{12}{}^{\alpha} + h''_{12}{}^{\beta}\right) - \left(h''_{11}{}^{\alpha} + h''_{11}{}^{\beta} + R_S\right)\left(h''_{22}{}^{\alpha} + h''_{22}{}^{\beta} + G_L\right)}$$

Analisi del secondo termine

□ Analisi del secondo termine

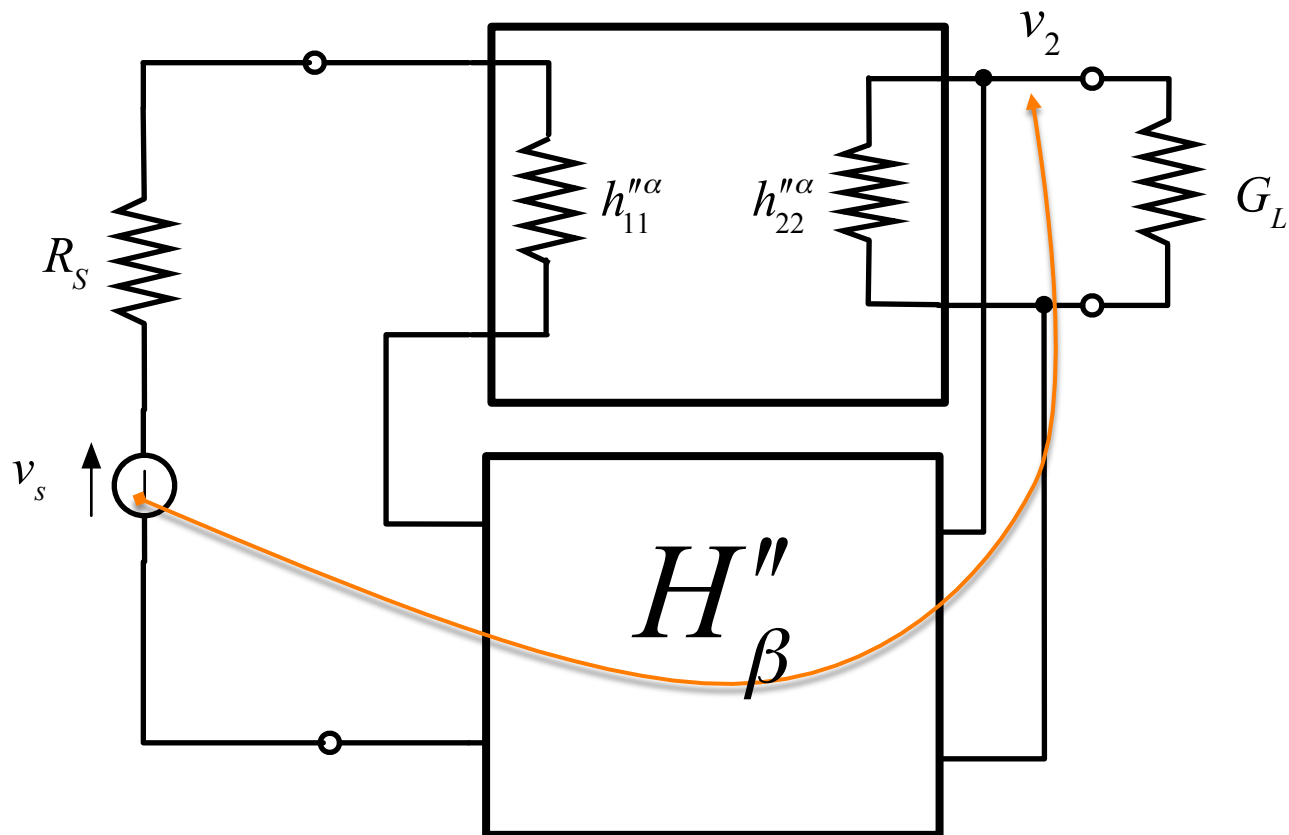
Guadagno diretto

$$A_{VP} = \frac{h_{21}''^{\beta}}{h_{21}''^{\beta}(h_{12}''^{\alpha} + h_{12}''^{\beta}) - (h_{11}''^{\alpha} + h_{11}''^{\beta} + R_s)(h_{22}''^{\alpha} + h_{22}''^{\beta} + G_L)} \frac{1}{1 - G_{loop}}$$

$$G_{diretto} = \frac{h_{21}''^{\beta}}{h_{21}''^{\beta}(h_{12}''^{\alpha} + h_{12}''^{\beta}) - (h_{11}''^{\alpha} + h_{11}''^{\beta} + R_s)(h_{22}''^{\alpha} + h_{22}''^{\beta} + G_L)}$$

$$A_{VP} = G_{diretto} \frac{1}{1 - G_{loop}}$$

G_{diretto} nel circuito



$$G_{\text{diretto}} = \frac{v_2}{v_s}$$

Analisi del secondo trasferimento

□ Impedenza di ingresso

$$\frac{1}{\det(H''_\gamma)} \begin{bmatrix} h''_{22} + h''_{22} + G_L & -(h''_{12} + h''_{12}) \\ -(h''_{21} + h''_{21}) & h''_{11} + h''_{11} + R_S \end{bmatrix} \begin{bmatrix} v_S \\ 0 \end{bmatrix} = \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

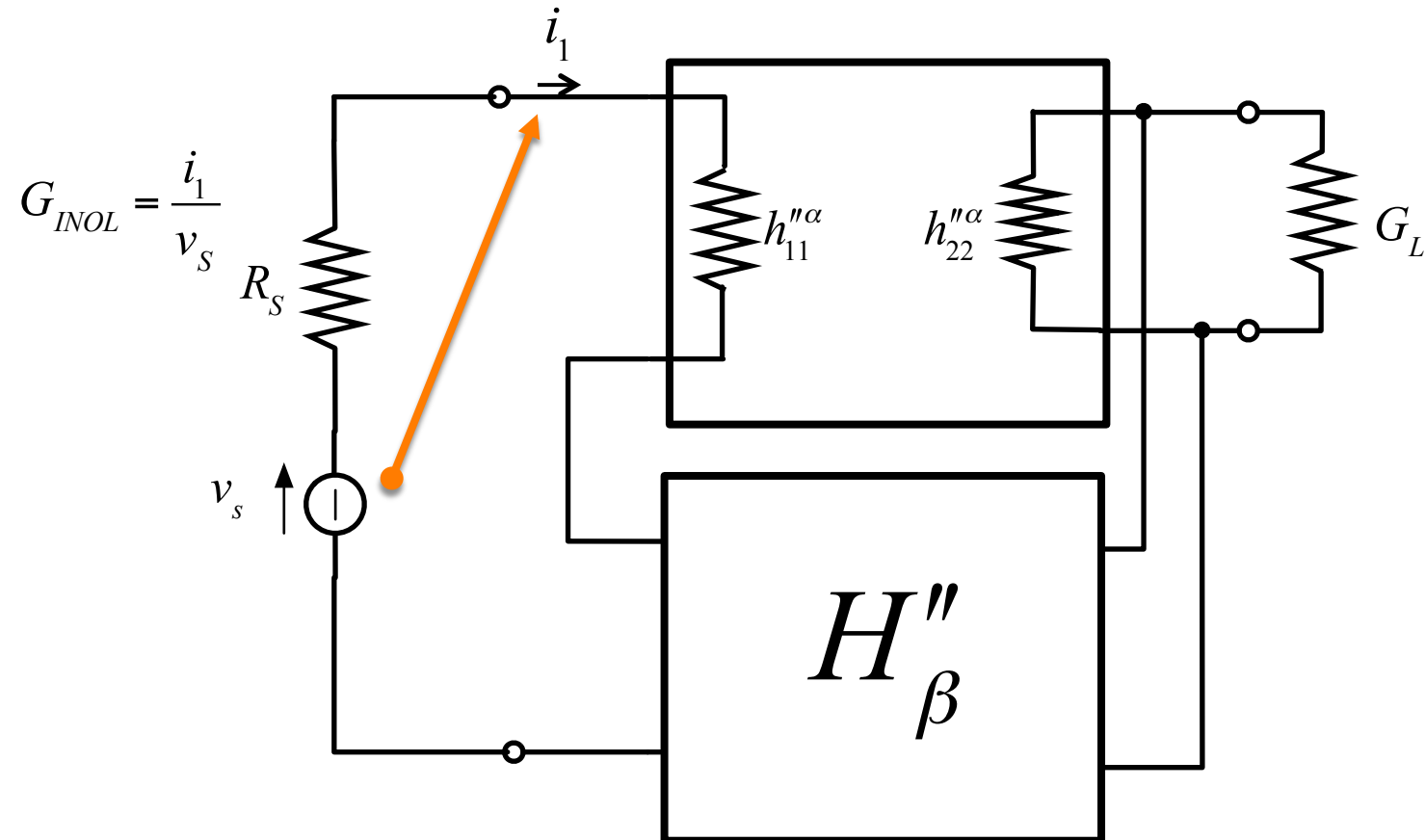
$$\frac{i_1}{v_S} = \frac{(h''_{22} + h''_{22} + G_L)}{\det H''_\gamma} = \frac{h''_{22} + h''_{22} + G_L}{(h''_{11} + h''_{11} + R_S)(h''_{22} + h''_{22} + G_L) - (h''_{12} + h''_{12})(h''_{21} + h''_{21})}$$

$$G_{IN} = \frac{h''_{22} + h''_{22} + G_L}{(h''_{11} + h''_{11} + R_S)(h''_{22} + h''_{22} + G_L) - (h''_{12} + h''_{12})h''_{21}} \cdot \frac{1}{1 - \frac{(h''_{12} + h''_{12})h''_{21}}{(h''_{11} + h''_{11} + R_S)(h''_{22} + h''_{22} + G_L) - (h''_{12} + h''_{12})h''_{21}}}$$

Conduttanza anello aperto

$$G_{IN} = G_{INOL} \frac{1}{1 - G_{loop}}$$

G_{IN} ad anello aperto nel circuito



Generazione delle varie combinazioni sullo stesso circuito

